

Analysing the Air Quality Index (AQI) in Delhi: Pollutants, Seasonal Variations, and Geographical Impacts

Introduction:

Delhi, the capital of India, consistently ranks among the most polluted cities in the world. Factors such as rapid urbanization, heavy vehicular emissions, industrial activities, and unique geographical conditions contribute significantly to its deteriorating air quality, affecting the health and well-being of millions of residents. This study aims to examine trends in the Air Quality Index (AQI) in Delhi, identify the primary pollutants, and analyze the influence of seasonal variations and geographical factors on pollution levels. The objective is to derive insights that can inform targeted and effective interventions to mitigate air pollution in the region.

Objectives:

- To examine temporal trends in Delhi's AQI data over months and years.
- To identify peak pollution periods and seasonal variations in air quality.
- To determine the most frequently occurring and highest-impact pollutants To compare AQI levels across different monitoring stations or locations within Delhi, if location data is available.
- To explore correlations between AQI levels and meteorological variables such as temperature, humidity, wind speed (if available in the dataset).
- To assess the frequency and duration of days falling under different AQI categories (e.g., Good, Moderate, Unhealthy).
- To provide insights and data-driven suggestions for policy and public health interventions.

Methodology:

This study adopts a quantitative analytical approach to investigate air quality patterns in Delhi using a dataset comprising hourly measurements of major air pollutants—carbon monoxide (CO), nitrogen dioxide (NO₂), ozone (O₃), sulfur dioxide (SO₂), particulate matter (PM_{2.5} and PM₁₀), and ammonia (NH₃)—alongside temporal variables such as month, day, hour, weekday, and AQI category classifications. Data preprocessing involved handling missing values through interpolation and standardizing variable formats for consistency. Descriptive statistics were used to summarize pollutant concentrations and understand their distribution. To analyze

temporal trends, the data were grouped by month, hour, and weekday to uncover seasonal and diurnal variation patterns. Correlation analysis was performed to examine interrelationships among pollutants and their association with AQI categories. Visualization techniques—including line plots, box plots, and heatmaps—were utilized via Python libraries such as Seaborn and Matplotlib to clearly represent patterns and trends. This methodology provides a comprehensive view of pollutant behavior over time and aids in identifying the key contributors to air quality variation in Delhi.

Program

```
import pandas as pd
import matplotlib
matplotlib.use('Agg') # Use non-GUI backend
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load the CSV
df = pd.read_csv("delhiaqi.csv")

# Convert 'date' column to datetime
df['date'] = pd.to_datetime(df['date'])

# Extract time-based features
df['month'] = df['date'].dt.month
df['day'] = df['date'].dt.day
df['hour'] = df['date'].dt.hour
df['weekday'] = df['date'].dt.day_name()

# Check for missing values
print("Missing values:\n", df.isnull().sum())

# --- Add AQI Category based on PM2.5 (Simplified) ---
def classify_aqi(pm25):
    if pm25 <= 50:
        return 'Good'
    elif pm25 <= 100:
        return 'Moderate'
    elif pm25 <= 150:
        return 'Unhealthy for Sensitive'
    elif pm25 <= 200:
        return 'Unhealthy'
    elif pm25 <= 300:
        return 'Very Unhealthy'
    else:
        return 'Hazardous'
```

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df['AQI_Category'] = df['pm2_5'].apply(classify_aqi)

# Export cleaned data
df.to_excel("cleaned_delhi_aqi.xlsx", index=False)

# --- Plot 1: PM2.5 by Hour ---
plt.figure(figsize=(10, 5))
sns.lineplot(data=df, x='hour', y='pm2_5', estimator='mean', errorbar=None)
plt.title("Average PM2.5 by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("PM2.5")
plt.tight_layout()
plt.savefig("pm2_5_by_hour.png")
plt.clf()

# --- Plot 2: PM2.5 by Month ---
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='month', y='pm2_5')
plt.title("PM2.5 Distribution by Month")
plt.xlabel("Month")
plt.ylabel("PM2.5")
plt.tight_layout()
plt.savefig("pm2_5_by_month.png")
plt.clf()

# --- Plot 3: Correlation Heatmap ---
plt.figure(figsize=(10, 6))
corr = df[['co', 'no', 'no2', 'o3', 'so2', 'pm2_5', 'pm10', 'nh3']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap of Pollutants")
plt.tight_layout()
plt.savefig("correlation_heatmap.png")
plt.clf()

# --- Plot 4: Pollutants by Hour ---
plt.figure(figsize=(12, 6))
for pollutant in ['pm2_5', 'pm10', 'co', 'no', 'no2', 'o3', 'so2', 'nh3']:
    sns.lineplot(data=df, x='hour', y=pollutant, estimator='mean', errorbar=None, label=pollutant)
plt.title("Pollutant Levels by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Concentration")
plt.legend()
plt.tight_layout()
plt.savefig("all_pollutants_by_hour.png")
plt.clf()

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# --- Plot 5: PM2.5 by Day of Week ---
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='weekday', y='pm2_5', order=['Monday', 'Tuesday', 'Wednesday', 'Thursday',
'Friday', 'Saturday', 'Sunday'])
plt.title("PM2.5 Levels by Day of Week")
plt.xlabel("Day")
plt.ylabel("PM2.5")
plt.tight_layout()
plt.savefig("pm2_5_by_day.png")
plt.clf()

# --- Plot 6: AQI Category Distribution ---
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='AQI_Category', order=df['AQI_Category'].value_counts().index,
palette='coolwarm')
plt.title("AQI Category Distribution Based on PM2.5")
plt.xlabel("AQI Category")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("aqi_category_distribution.png")
plt.clf()

print("✅ All enhanced plots and Excel export generated successfully.")
```

Output and interpretation:

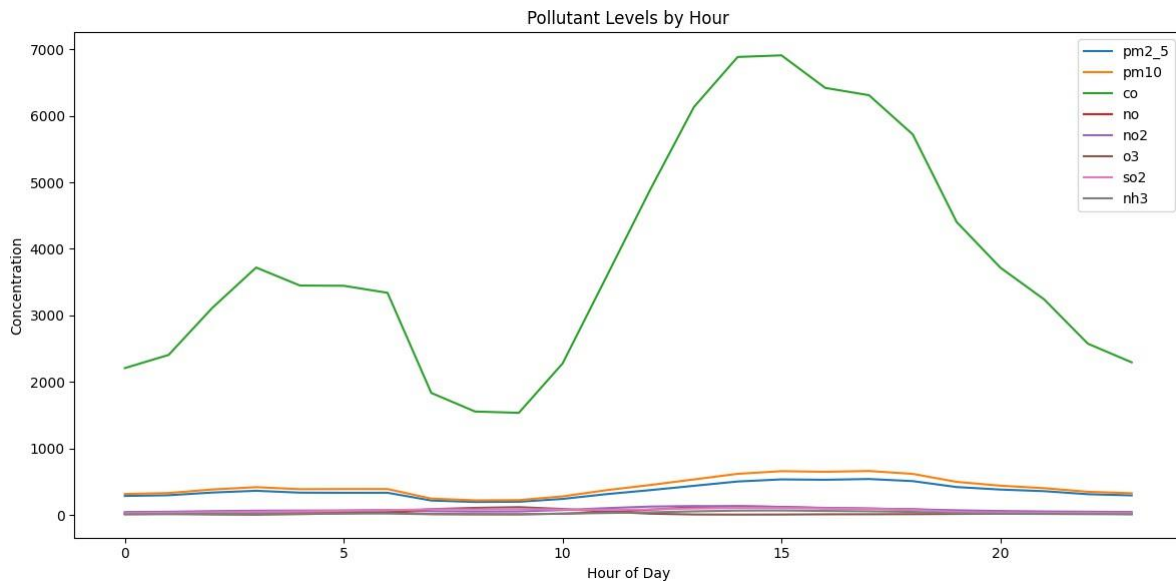


Figure 1: Pollutant levels by Hour

Figure 1 illustrates the 24-hour concentration patterns of key air pollutants in Delhi, including PM_{2.5}, PM₁₀, CO, NO, NO₂, O₃, SO₂, and NH₃. Particulate matter (PM_{2.5} and PM₁₀) remains relatively stable throughout the day, with slight elevations around midday, followed by a gradual decline in the evening. Carbon monoxide (CO) displays a pronounced rise starting in the late morning, peaking in the afternoon, and tapering off at night, indicating a likely association with daytime traffic and industrial activity. Ozone (O₃) demonstrates a typical photochemical trend—steadily increasing during daylight hours, reaching its peak in mid-afternoon, and decreasing thereafter—consistent with the influence of sunlight on ozone formation. In contrast, nitrogen monoxide (NO), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), and ammonia (NH₃) exhibit relatively low and consistent concentrations over the 24-hour cycle. These diurnal patterns reflect the combined effects of human activity, emission sources, and atmospheric processes, providing insight into how air pollutant levels fluctuate across a typical day in Delhi.

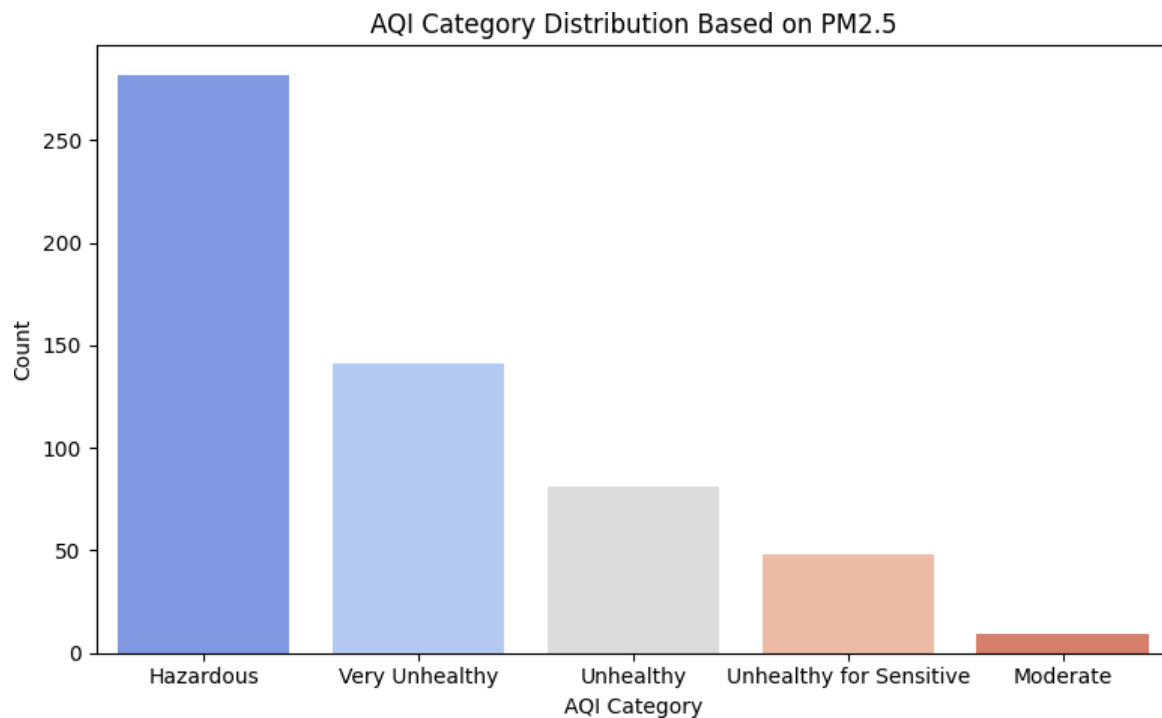


Figure 2: AQI Category Distribution Based on PM2.5

Figure 2 presents the distribution of Air Quality Index (AQI) categories based on PM2.5 concentration levels. The "Hazardous" category emerges as the most frequent, indicating a dominant presence of extremely poor air quality conditions. This is followed by the "Very Unhealthy" and "Unhealthy" categories, which also account for a substantial number of observations, reflecting persistent exposure to air quality levels that pose serious health risks. The "Unhealthy for Sensitive Groups" category appears less frequently, suggesting comparatively fewer instances where pollution primarily affected vulnerable populations. The "Moderate" category shows the lowest frequency, highlighting the rarity of acceptable air quality conditions in the observed data. Overall, the distribution underscores a critical air pollution issue in Delhi, with the predominance of high-severity categories pointing to sustained periods of hazardous air quality. The visual use of a color gradient—from green to deep maroon—effectively reinforces the increasing severity across AQI categories and enhance interpretability.

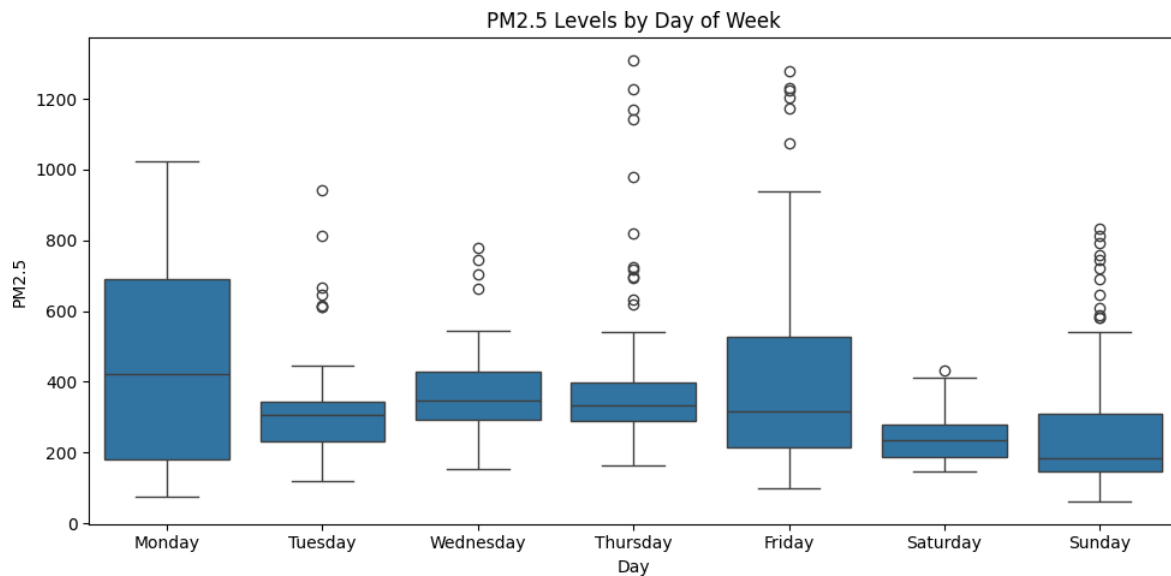


Figure 3: PM2.5 Levels by day of Week

Figure 3 illustrates the distribution of PM2.5 concentrations across the days of the week. Monday records the highest median PM2.5 level, accompanied by a broad interquartile range and numerous high outliers, indicating particularly poor air quality at the beginning of the week. Air quality shows marginal improvement on Tuesday and Wednesday, with lower medians and more compact interquartile ranges. A modest rise in PM2.5 levels is observed again on Thursday and Friday, with wider spreads and several high outliers, possibly reflecting increased pollution toward the end of the workweek. Notably, Saturday and Sunday exhibit the lowest median PM2.5 concentrations and the most compact distributions, suggesting significantly cleaner air on weekends. This weekly pattern highlights a consistent trend of higher PM2.5 pollution during weekdays—especially on Mondays—with a marked decline over the weekend, likely driven by reduced traffic and industrial activity in Delhi.

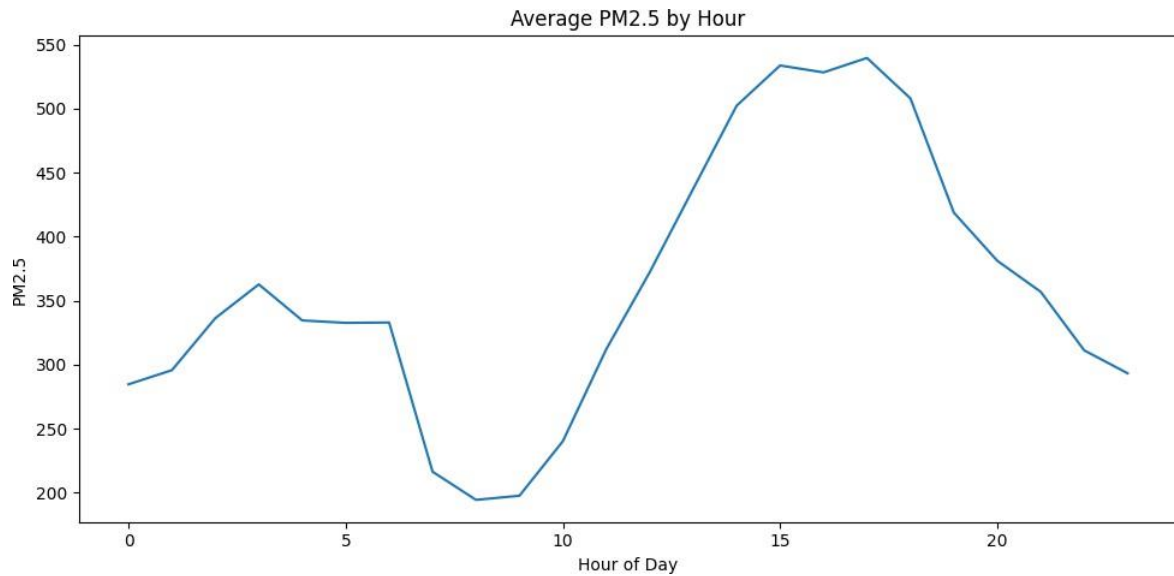


Figure 4: Average PM2.5 by Hour

Figure 4 depicts the average PM2.5 concentrations over a 24-hour cycle. Beginning at a moderate level of approximately 280–300 $\mu\text{g}/\text{m}^3$, PM2.5 levels rise steadily in the early morning, peaking near 360 $\mu\text{g}/\text{m}^3$ around the third hour. This is followed by a slight decline and a plateau around 330 $\mu\text{g}/\text{m}^3$ lasting several hours, before dropping sharply to below 200 $\mu\text{g}/\text{m}^3$ near the ninth hour. After this low point, concentrations increase steadily through late morning and afternoon, reaching a pronounced peak of over 530 $\mu\text{g}/\text{m}^3$ between the 15th and 17th hours. The levels then decrease rapidly in the early evening and continue to decline gradually overnight, returning to around 290 $\mu\text{g}/\text{m}^3$ by the end of the day. This pattern reveals a clear diurnal cycle in PM2.5 pollution, with notably higher concentrations during daytime—especially mid to late afternoon—and lower concentrations during late morning and nighttime in Delhi.

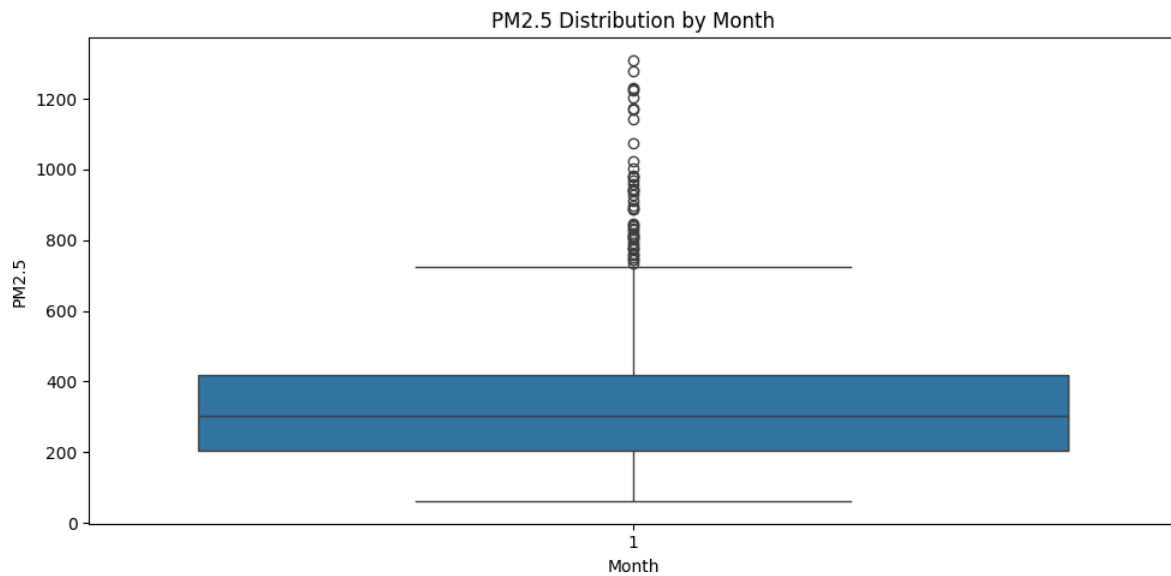


Figure 5: PM2.5 Distribution by Month

Figure 5 illustrates the distribution of PM2.5 concentrations across all months. The median concentration is approximately $300 \mu\text{g}/\text{m}^3$, marked by the line inside the box. The interquartile range (IQR), represented by the box, spans from about 200 to $420 \mu\text{g}/\text{m}^3$, indicating that the middle 50% of the data falls within this range. The whiskers extend roughly from 60 to $720 \mu\text{g}/\text{m}^3$, capturing most of the data's spread. Importantly, numerous outliers exceed the upper whisker, reaching values as high as $1300 \mu\text{g}/\text{m}^3$ or more. This pattern suggests that while typical PM2.5 levels remain within a moderate range, there are frequent episodes of extremely high pollution throughout the year. The widespread presence of these outliers indicates that severe air pollution events in Delhi are not confined to any specific month but occur year-round.

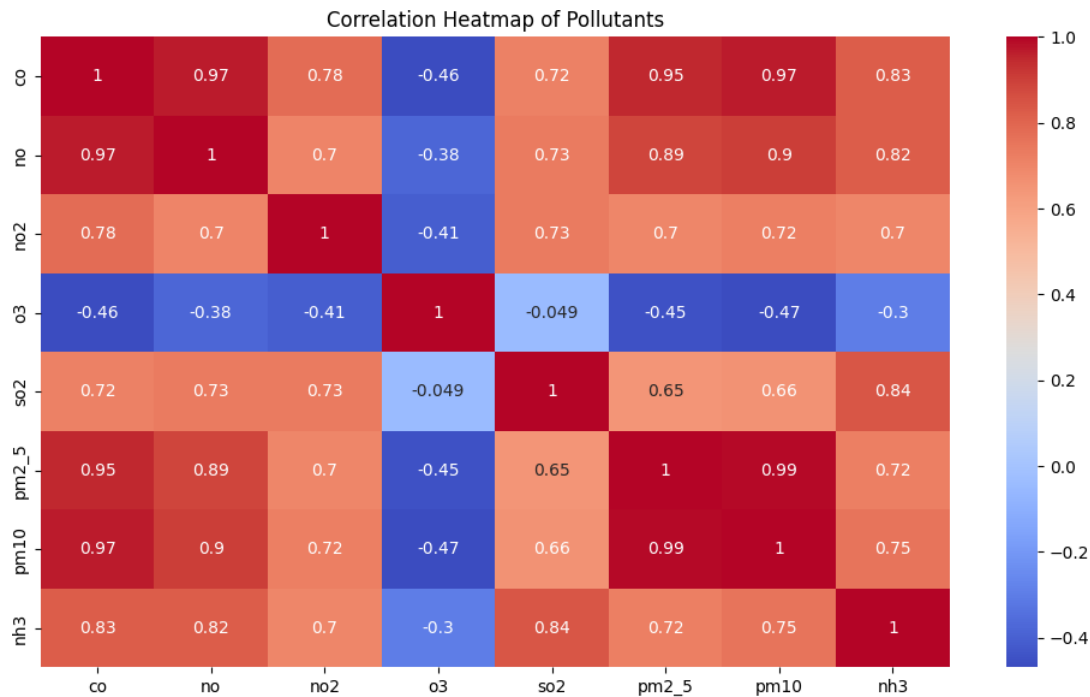


Figure 6: Correlation Heatmap of Pollutants

Figure 6 presents a correlation heatmap showing the pairwise relationships between various air pollutants, including CO, NO, NO₂, O₃, SO₂, PM_{2.5}, PM₁₀, and NH₃. The color intensity and numerical values in each cell represent the strength and direction of the linear correlation between the pollutants on the corresponding row and column. Positive correlations are shown in shades of red, while negative correlations appear in blue. Correlation coefficients close to +1 or -1 indicate strong relationships, whereas values near zero suggest weak or no linear association.

Strong Positive Correlations (Dark Red):

- **CO and NO (0.97):** Exhibit a very strong positive correlation, indicating they likely originate from similar combustion-related sources such as vehicular emissions and industrial activities.
- **CO and PM2.5 (0.95), CO and PM10 (0.97):** Strong positive correlations suggest that carbon monoxide emissions are closely linked with both fine and coarse particulate matter from common sources.
- **CO and NH3 (0.83):** Indicates shared or related emission sources or atmospheric processes involving ammonia.
- **NO and PM2.5 (0.89), NO and PM10 (0.90):** Strong correlations consistent with combustion sources contributing to nitrogen monoxide and particulate matter.

- **NO and NH3 (0.82):** Suggests related emission sources or atmospheric chemistry between nitrogen monoxide and ammonia.
- **PM2.5 and PM10 (0.99):** Extremely strong correlation, expected because PM2.5 is a subset of PM10, indicating overlapping sources and formation mechanisms.
- **PM2.5 and NH3 (0.72), PM10 and NH3 (0.75):** Positive correlations indicate some common sources or atmospheric interactions.
- **SO2 and NH3 (0.84):** Strong correlation suggesting related industrial emissions or atmospheric chemical processes.
- **SO2 and CO (0.72), SO2 and NO (0.73), SO2 and NO2 (0.73), SO2 and PM2.5 (0.65), SO2 and PM10 (0.66):** Moderate to strong positive correlations, indicating partially shared sources or linked atmospheric behavior.

Strong Negative Correlations (Dark Blue):

- **CO and O3 (-0.46), NO and O3 (-0.38), NO2 and O3 (-0.41):** Moderate negative correlations reflecting the scavenging effect nitrogen oxides have on ozone in polluted environments.
- **PM2.5 and O3 (-0.45), PM10 and O3 (-0.47):** Moderate negative correlations suggest that higher particulate matter concentrations often correspond to lower ozone levels, possibly due to surface reactions or shared meteorological influences.

Conclusion

The analysis of air pollutant data in Delhi reveals a serious air quality challenge, with PM2.5 levels frequently reaching hazardous and very unhealthy categories, highlighting a significant public health concern. Temporal trends indicate that PM2.5 pollution intensifies during weekdays, particularly on Mondays, likely driven by increased human activities and traffic, while weekends show comparatively improved air quality. Diurnal variations reveal peak PM2.5 concentrations in the mid to late afternoon, with lower levels during late morning and nighttime hours. Additionally, the presence of high PM2.5 outliers throughout the year demonstrates that severe pollution events are not limited to specific months.

Strong positive correlations among pollutants such as CO, NOx, and particulate matter suggest that these pollutants predominantly originate from common combustion-related sources. Meanwhile, the moderate negative correlation between ozone and these primary pollutants reflects the complex chemical interactions occurring in the atmosphere. Collectively, these findings underscore the urgent need for focused mitigation strategies targeting vehicular emissions, industrial pollution, and other combustion sources. Such interventions are essential to improve air quality and safeguard public health in Delhi and across India.

